

Paper 1 - Set 01 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	D	Ratio, proportion and percentages
2	C	Number, units and estimation
3	F	Exponentials and logarithms
4	A	Exponentials and logarithms
5	E	Circles and circle theorems
6	A	Binomial expansion
7	B	Sequences and series
8	B	Indices and surds
9	E	Integration
10	A	Simultaneous equations and inequalities
11	C	Trigonometric functions and equations
12	A	GCSE algebra
13	F	Triangles, radians and sectors
14	A	Simultaneous equations and inequalities
15	B	Sequences and series
16	E	Binomial expansion
17	D	Straight lines
18	C	Graphs of functions
19	D	Indices and surds
20	A	Probability

Worked solutions

Question 1 RAT-06 · Ratio, proportion and percentages · spec M3.10

Two similar solids have surface areas in the ratio 4 : 9. Find the ratio of their volumes.

Answer: D

Length ratio 2 : 3, so volume ratio $2^3 : 3^3 = 8 : 27$.

Question 2 NUM-03 · Number, units and estimation · spec M2.9

Express the recurring decimal $0.\dot{2}7$ as a fraction in its lowest terms.

Answer: C

$x = 0.2727\dots$, $100x = 27.2727\dots$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$.

Question 3 LOG-13 · Exponentials and logarithms · spec MM5.2

Given that $\log_{10} 2 = p$ and $\log_{10} 3 = q$, express $\log_{10} 0.75$ in terms of p and q .

Answer: F

$0.75 = \frac{3}{4}$, so $\log 0.75 = \log 3 - \log 4 = q - 2p$.

Question 4 LOG-10 · Exponentials and logarithms · spec MM5.3

How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?

Answer: A

$y = 2^x$: $y^2 + 2y - 8 = (y + 4)(y - 2) = 0$. $y = -4$ is impossible, $y = 2$ gives $x = 1$. One solution.

Question 5 CIR-14 · Circles and circle theorems · spec MM3.3

AB is a diameter of a circle and C is a point on the circle with $AC = 6$ and $BC = 8$. Find the radius of the circle.

Answer: E

The angle in a semicircle is a right angle, so $AB = \sqrt{36 + 64} = 10$ and the radius is 5.

Question 6 BIN-02 · Binomial expansion · spec MM2.4

The coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60. Given that $k > 0$, find k .

Answer: A

$\binom{6}{2}k^2 = 15k^2 = 60$, so $k^2 = 4$ and $k = 2$.

Question 7 SEQ-14 · Sequences and series · spec MM2.1

The sequence $u_n = n^2 - 10n + 30$ is defined for positive integers n . What is the smallest value taken by any term of the sequence?

Answer: B

$u_n = (n - 5)^2 + 5$, minimised at $n = 5$ with value 5.

Question 8 IND-08 · Indices and surds · spec MM1.1

Solve $9^x - 4 \times 3^x + 3 = 0$.

Answer: B

Let $y = 3^x$: $y^2 - 4y + 3 = (y - 1)(y - 3) = 0$, so $3^x = 1$ or $3^x = 3$, giving $x = 0$ or $x = 1$.

Question 9 INT-13 · Integration · spec MM7.3

$f(x) = ax + b$ satisfies $\int_0^1 f(x) dx = 2$ and $\int_0^2 f(x) dx = 6$. Find $f(1)$.

Answer: E

$\frac{a}{2} + b = 2$ and $2a + 2b = 6$, so $a + b = 3$; subtracting the first equation from this gives $\frac{a}{2} = 1$, so $a = 2$, $b = 1$ and $f(1) = 3$.

Question 10 SIM-05 · Simultaneous equations and inequalities · spec MM1.5

For how many integers n is $n^2 - 10n + 16 < 0$?

Answer: A

$(n - 2)(n - 8) < 0 \Rightarrow 2 < n < 8$, so $n = 3, 4, 5, 6, 7$: five integers.

Question 11 TRG-18 · Trigonometric functions and equations · spec MM4.3

Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

Answer: C

$\tan 30^\circ \tan 60^\circ = \frac{1}{\sqrt{3}} \cdot \sqrt{3} = 1$, $\sin^2 45^\circ = \frac{1}{2}$. Total $\frac{3}{2}$.

Question 12 ALG-06 · GCSE algebra · spec M4.3

Given that $v^2 = u^2 + 2as$, find s when $v = 13$, $u = 5$ and $a = 4$.

Answer: A

$169 = 25 + 8s \Rightarrow s = 18$.

Question 13 TRI-01 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 5$, $AC = 8$ and angle $BAC = 60^\circ$. Find BC .

Answer: F

Cosine rule: $BC^2 = 25 + 64 - 2 \cdot 5 \cdot 8 \cdot \frac{1}{2} = 89 - 40 = 49$, so $BC = 7$.

Question 14 SIM-02 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $2x^2 - 5x - 3 \geq 0$.

Answer: A

$(2x + 1)(x - 3) \geq 0$: the roots are $-\frac{1}{2}$ and 3 and the parabola opens upwards, so $x \leq -\frac{1}{2}$ or $x \geq 3$.

Question 15 SEQ-05 · Sequences and series · spec MM2.3

A geometric series has first term a and common ratio r with $|r| < 1$. Its sum to infinity is 8 and the sum to infinity of the squares of its terms is $\frac{64}{3}$. Find r .

Answer: B

$$\frac{a}{1-r} = 8 \text{ and } \frac{a^2}{1-r^2} = \frac{64}{3}. \text{ Dividing the second by the square of the first: } \frac{1-r}{1+r} = \frac{64/3}{64} = \frac{1}{3}, \text{ so } 3-3r=1+r \text{ and } r=\frac{1}{2}.$$

Question 16 BIN-03 · Binomial expansion · spec MM2.4

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

Answer: E

$$\text{General term } \binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-3r}. \text{ Need } 12-3r=0, r=4: \binom{6}{4} \times 16 = 15 \times 16 = 240.$$

Question 17 LIN-03 · Straight lines · spec MM3.1

The line L passes through $(4, 0)$ and $(0, 6)$. Find the perpendicular distance from the origin to L .

Answer: D

$$\text{Area of triangle} = \frac{1}{2} \cdot 4 \cdot 6 = 12. \text{ Hypotenuse} = \sqrt{16+36} = \sqrt{52} = 2\sqrt{13}. \text{ Distance } d \text{ satisfies } \frac{1}{2} \cdot 2\sqrt{13} \cdot d = 12, \text{ so } d = \frac{12}{\sqrt{13}}.$$

Question 18 GRA-07 · Graphs of functions · spec MM8.5

The graph of $y=f(x)$ has $f'(x) > 0$ for $x < 1$, $f'(1) = 0$, $f'(x) > 0$ for $1 < x < 3$, $f'(x) < 0$ for $x > 3$. Which of the following is true?

Answer: C

At $x = 1$ the derivative is zero but does not change sign: a stationary point that is not a turning point (an inflexion).
At $x = 3$ the derivative changes from positive to negative: local maximum.

Question 19 IND-09 · Indices and surds · spec MM1.2

Find the value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{2+\sqrt{3}}$.

Answer: D

Rationalising each term gives $(\sqrt{2}-1) + (\sqrt{3}-\sqrt{2}) + (2-\sqrt{3})$, which telescopes to 1.

Question 20 PRO-08 · Probability · spec M7.7

A box contains n white balls and 3 black balls. Two balls are drawn without replacement. The probability that both are white is $\frac{5}{12}$. Find n .

Answer: A

$$\frac{n}{n+3} \cdot \frac{n-1}{n+2} = \frac{5}{12} \Rightarrow 12n^2 - 12n = 5n^2 + 25n + 30 \Rightarrow 7n^2 - 37n - 30 = 0 \Rightarrow (7n+5)(n-6) = 0, \text{ so } n = 6.$$